

Comparison of Explainable AI Techniques for Early Detection of Potential Prediction Failures

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Abstract—In this paper, the characteristics of explainability, such as the diversity and instability, of Explainable Artificial Intelligence (XAI) techniques have been studied using convolutional neural network (CNN) models and three types of data characteristics (natural, medical, and fashion images). The understanding of explainability characteristics of XAI techniques is essential for human interpretability of AI-explainability to detect prediction failures early at training by considering the highest confidence predictors. Therefore, a negation-based XAI technique (named nEGXAI-V) for CNN-models, as a variant of the recently introduced nEGXAI for tabular data, is proposed and compared with the traditional XAI frameworks, called Gradient-SHAP, Integrated Gradients, Superpixel LIME, Smooth Occlusion, and Smooth Grad-CAM techniques. The diversity of explainability of these XAI techniques is studied using VGG16, VGG19, and ResNet18 models while their instability of explainability is studied using the public datasets, namely Intel Image Classification data, HAM10000 Medical data, and Fashion-MNIST data. Experimental results show that the traditional XAI techniques suffer from explainability diversity and instability issues, leading to difficulties in interpreting AI explainability by human intelligence (HI). Results also show that nEGXAI-V, in contrast, utilizes characteristics of diversity and instability along with the concept of constructive negation to improve HI interpretability of AI explainability by detecting the contributing and competing features of class predictions and prediction failures through its non-affirmation logic by avoiding affirmation bias.

Index Terms—Explainable AI, convolutional neural networks, negation-based XAI, LIME, SHAP

I. INTRODUCTION

Trustworthy explainable AI is essential for many applications that include natural image classification, human-generated image recognition, and medical image analytics. Many explainable AI (XAI) techniques, including Local Interpretable Model-agnostic Explanations (LIME [1]), SHapley Additive exPlanations (SHAP [2]), and Gradient-weighted Class Activation Mapping (Grad-CAM [3]), have been proposed in AI literature to address the problems of explainability in CNN-based models. However, their trustworthiness is still questionable because of the problems of diversity and instability of explainability to different models and data characteristics. To overcome these issues in XAI techniques, in-depth research continues with the goal of achieving trustworthy XAI technologies. To develop trustworthy XAI frameworks, it is important to focus on HI interpretability of AI explainability that can detect possible prediction failures early at training

while it can explain the reasoning for the predicted results and decisions. The early detection of potential prediction failures must come from the HI interpretability of AI explainability at training phases of the development of deep learning (DL) models (e.g., CNN-based models). In essence, this should come from the confidence levels of deep learning models on their predictions at training. Therefore, the methods proposed in this paper focus on the best positive confidence and the negative (competing class) confidence. Most recently, a negation-based explainable AI technique (called nEGXAI [4]) was proposed for tabular data; however, nEGXAI-based XAI techniques for visual data (e.g., images or video) are yet to be developed.

The goal of this paper is to first develop nEGXAI-V for visual data and then compare the performance of explainability behavior of Gradient-SHAP, Integrated Gradients, Superpixel LIME, Smooth Occlusion, Smooth Grad-CAM, and nEGXAI-V techniques when applied to (i) VGG16 [5], VGG19 [5], and ResNet18 [6] CNN-based models that bring the issue of diversity of explainability and (ii) Intel Image Classification data [7], HAM10000 Medical data [8], and Fashion-MNIST data [9] that bring the issues of instability of explainability through complex and heterogeneous data characteristics.

II. METHODS

In this study three CNN-based models, three types of datasets, and six AI explainers are used to demonstrate that HI interpretability of AI explainability can be developed by leveraging the diversity and instability properties of XAI approaches. Also, the explainability results of XAI techniques are fused to generate a benchmark explainability to measure the effects of diversity and instability of explainability of XAI techniques. It is important to note that, in the proposed methodology, early detection of potential prediction failures is performed at the training phase of DL models by understanding the contributions of the highest confidence positive and negative (or competing) predictors of CNN-based models.

A. CNN-based Models

In this study, VGG16, VGG19, and ResNet18 are used for understanding the explainability behavior of XAI techniques based on the diversity properties developed through fast, detailed, and organized AI learning models [10]. VGG16 is one of the CNN-based models considered in this approach. It is

defined as a fast learner model in a recent paper [10]. VGG16 is proposed by Oxford's Visual Geometric Group (VGG) and it consists of 16-layer convolutional neural network architecture, where 13 layers are convolutional layers and 3 layers are fully connected layers. VGG19 is the second model considered in this methodology. It was described as detailed learner model in a recent work [10]. VGG19 was also developed by Oxford's Visual Geometric Group. This has 16 convolutional layers and 3 fully connected layers, making it a detailed learner.

ResNet18 is also considered in the proposed empirical study. It is an organized learning model. It was proposed by Microsoft Research, and it acts like an organized learning model and uses a skip-connections concept to avoid vanishing gradient problem. The differences in model architecture and intelligence logic, the application of XAI techniques to these models generate significant diversity of explainability. Hence, these models are suitable to adopt and analyze explainability behaviors of XAI techniques with respect to CNN-models.

B. Traditional XAI Techniques

The XAI approaches considered are the Gradient-SHAP, Integrated Gradients, Superpixel LIME, Smooth Occlusion, Smooth Grad-CAM and nEGXAI-V, where nEGXAI-V is a variant of the original nEGXAI that was proposed recently for tabular data. We can rename the tabular nEGXAI by nEGXAI-T for easy reference. Gradient-SHAP is considered in this experimental study since it is specifically designed for deep learning techniques that include CNN-models by combining both SHAP values and Integrated Gradient, enabling the detection of important features that contribute to explainability.

Integrated Gradient XAI technique is also considered since it can provide the rate of change of CNN models prediction by associating them with the background effects of an image. Hence, it follows the way humans interpret explainability. Superpixel LIME model is used since it can efficiently localize the explainability; thus, it can support human interpretability by localizing the explainability to the features belonging to small segments of images. In other words, this explainer identifies the local explainers (small region of features) that contribute to the changes experienced by models' prediction.

Smooth occlusion is a sensitivity-based interpretability approach. It applies smooth perturbations to the local regions of an image and measures model's predictive responses to the change of perturbation levels and regions. Smooth Grad-CAM uses high-order derivatives that help localize object details and SmoothGrad technique that helps smooth noise along the edge by averaging edge gradients. It facilitates visual interpretability and prioritizes its application to CNN-based models.

Additionally, nEGXAI-V, as a variant of nEGXAI, is proposed for CNN-based models and compared with other XAI techniques. The selected XAI techniques provide diverse configurations; hence, stability of interpretability can be studied.

C. Negation-Based XAI

This section presents the algorithm for the proposed nEGXAI-V. Suppose we have a model f that is trained on a

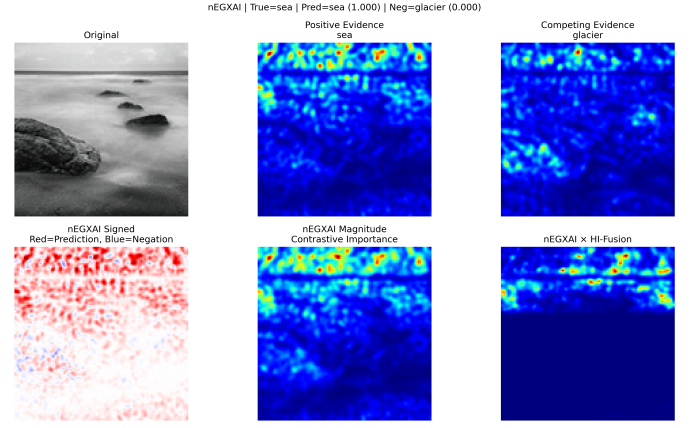


Fig. 1. Sea image from Intel Image Classification data and nEGXAI-V results

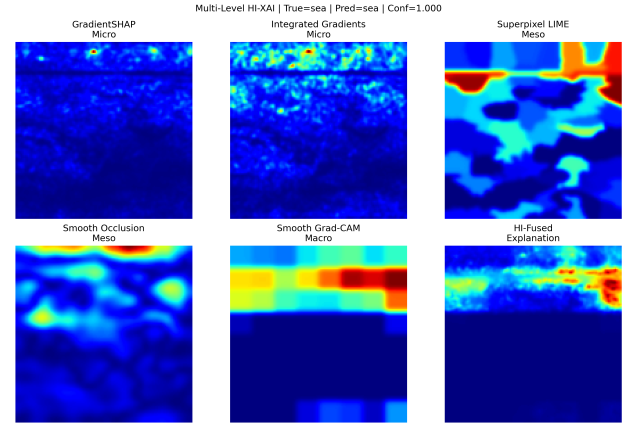


Fig. 2. XAI results of Sea image from Intel Image Classification dataset

feature space F and x is an image in F , c_p is its predicted class with the highest positive confidence, and c_n is its negative class that is determined based on its strongest competing class. The positive and negative classes are determined based on their probabilities calculated by using a softmax function.

Suppose g_p and g_n are integrated gradients calculated from (x, c_p) and (x, c_n) , respectively. These gradients reflect both positive and negative gradient values, indicating the proposed nEGXAI-V considers both the positive and negative (competing) classes; hence, defines the evidence that supports the predicted class (S_p) and the evidence that supports the strongest competing class (S_n) respectively as follows:

$$S_p = \max(g_p, 0) \text{ and } S_n = \max(g_n, 0). \quad (1)$$

As such the negation-based contrast (concept of constructive negation) is assumed to be the measure of explainability:

$$\rho = S_p - S_n, \quad (2)$$

and it is split into signed and magnitude explanations vectors:

$$\rho_m = |\rho| \text{ and } \rho_s = \rho / \max(\rho - \rho_m), \quad (3)$$

where ρ_m highlights the strongest contrastive regions and ρ_s is the direction as the measure to highlight the regions for the

positive and negative classes; thus, the use of contributors to the prediction of the positive class can be explained.

The main concept of nEGXAI is to utilize constructive negation (features negating predicted class) in the Fourier domain for detecting possible prediction failures based on non-contributing features to develop trustworthy and interpretable decision [4]. However, the variant proposed in this paper for CNN-based models considers constructive negation in non-Fourier domain (i.e., in spatial domain). The distinction is that the traditional XAI approaches shows the features that are directly contributing to the predicted class, but nEGXAI-V, based on its non-affirmation logic, shows both positive and competing (negative) features to avoid affirmation bias.

III. EMPIRICAL RESULTS

The purpose of this experiment is to apply the proposed nEGXAI-V and other XAI methods to VGG16, VGG19, and ResNet18 models that are trained on the Intel Image Classification, Fashion-MNIST, and HAM100000 datasets, generate explainability results under the constraints of diversity and instability of explainability, and interpret them. In essence, it is to show the presence of diversity and instability of explainability in the output of the traditional XAI techniques. It is also to show that the proposed nEGXAI-V reduces these problems and enhances HI interpretability of AI explainability at training by detecting potential prediction failures.

A. Validation of XAI techniques

In this experiment, the traditional XAI techniques (GradientSHAP, Integrated Gradients, Superpixel LIME, Smooth Occlusion, Smooth Grad-CAM techniques) and nEGXAI-V are compared. Figs. 1 and 2 show the explainability results of the XAI techniques using the sea (natural) image. Overall observation is that the traditional XAI techniques detect positive evidences associated with the border between the sea and the sky as contributing features of the high confidence prediction by VGG16. The last image in Fig. 2 is the HI-fused explanation generated by fusion of the results of the traditional XAI techniques in the figure. Comparing the results of XAI techniques with the HI-fused explainer shows that each XAI technique has its unique combination of positive evidences. However, Fig. 1 results show that nEGXAI-V delivers additional knowledge that enhances HI-interpretability through positive evidences, negative evidence, contrastive importance, and nEGXAI-fused explanation. For example, the nEGXAI signed image in the figure highlights the positive evidence with red color and negative evidence by blue color to avoid affirmation bias in HI interpretability.

B. Instability of Explainability

Complexity and heterogeneity data characteristics of the Intel Image Classification, Fashion-MNIST, and HAM10000 datasets are utilized to study instability of explainability of XAI techniques. The first images in Figs. 1, 3, and 5 show the highest confidence predictors of VGG16 under the complexity

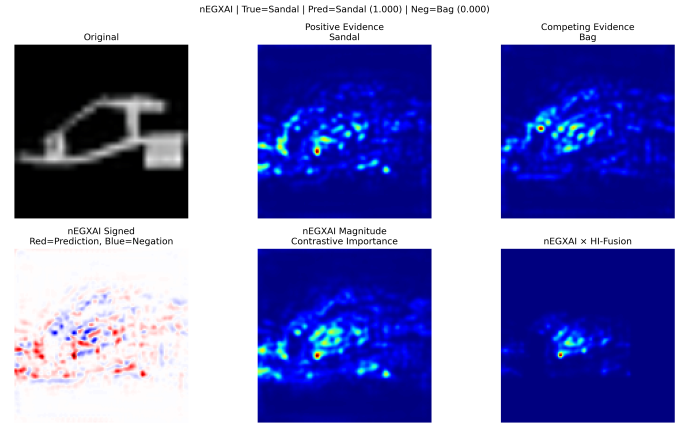


Fig. 3. Sandal image from Fashion-MNIST dataset and its nEGXAI-V results

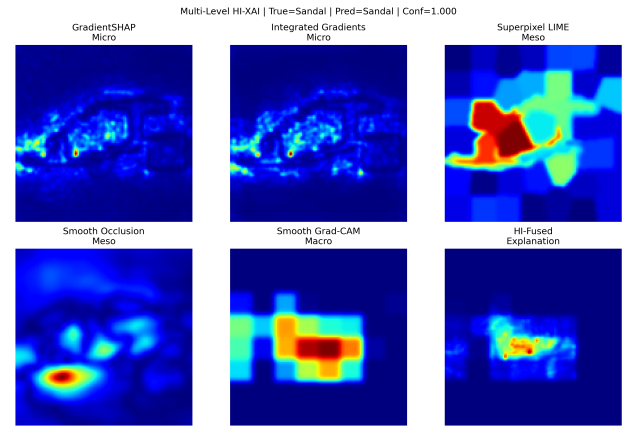


Fig. 4. XAI results of Sandal image from Fashion-MNIST dataset

and heterogeneity characteristics of the data types, respectively. The true labels of these images are sea, sandals, and Melanocytic nevi, respectively. Their predicted label with the highest confidence by VGG-16 are 1.00, 1.0, and 0.862. Now the goal is to understand the results of the explainers under these highest confidence situations such that HI interpretability can be established. Figs. 1, 3, and 5 show the results of nEGXAI-V explainer whereas Figs. 2, 4, and 6 show the results of the traditional XAI techniques and their HI-fused explanations, respectively. We can interpret these results as follows: the GradientSHAP and Integrated Gradient XAI techniques, explain VGG16 reasoning at *micro*-level (small-scale) by highlighting the locally contributing features of the sky or the sea region, whereas the Superpixel LIME and Smooth Occlusion highlights at *meso*-level (interaction between small and large-scale feature contributors) the border between the sea and the sky. Similarly, Smooth GradCAM highlights at *macro*-level by targeting the sea features. Finally, the fused explanation highlights the border and the sea details at the combined micro, meso and macro levels; thus, it facilitates HI interpretability. In the same order, the results for the sandals and Melanocytic nevi are presented in Figs. 5 and 6.

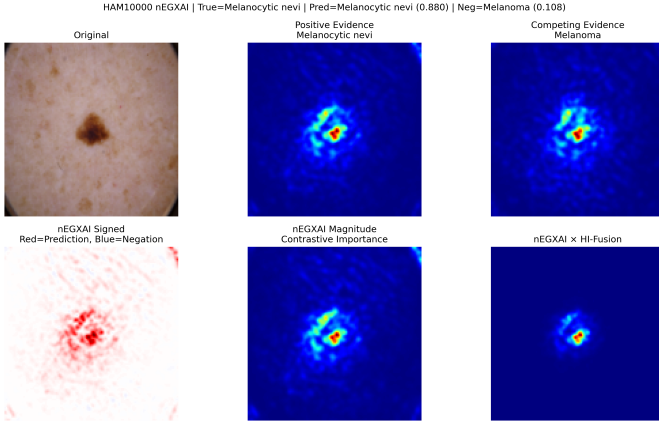


Fig. 5. nEGXAI-V results of Melanocytic nevi skin cancer from HAM10000

C. Diversity of Explainability

To study diversity of explainability, the same experiment is conducted with VGG19 and ResNet18, in addition to VGG16. The results showed similar patterns over data characteristics while producing distinct characteristics over the CNN-based models, indicating the presence of diversity of explainability.

D. Final Findings

The comprehensive analysis shows that XAI approaches have diversified-explainability and modality-instability over distinct data characteristics; however, they can be leveraged to generate better human interpretability. In summary:

- Gradient-SHAP and Integrated Gradient provides micro-level human interpretability.
- Superpixel LIME and Smooth Occlusion provides meso-level human interpretability.
- Smooth Grad-CAM provides macro-level interpretability.

They can be fused to give a balanced human interpretability over AI explainability. On the other hand, the proposed nEGXAI-V can provide stable AI explainability. It means that the proposed nEGXAI-V approach can alone provide positive evidence and competing evidence with the magnitude of contrastive importance without the need for fusion.

Also, in a separate experiment, mean deletion scores are calculated to quantify explainability. These scores of Gradient-SHAP, Integrated Gradients, Superpixel LIME, Smooth Occlusion, Smooth Grad-CAM, nEGXAI-V and nEGXAI-Fused models are 0.502, 0.458, 0.392, 0.333, -0.40, 0.576, and 0.509, respectively. The highest mean deletion score shows nEGXAI-V outperforms the other standard explainability models.

IV. CONCLUSION

This research analyzed multiple XAI approaches, CNN-based models, and datasets in terms of enhancing HI interpretability of AI explainability. The proposed experimental analysis showed the presence of diversity and instability in AI explainability that can create uncertainty in human interpretability; however, it can be reduced by integrating the concept of constructive negation that avoids affirmation-bias.

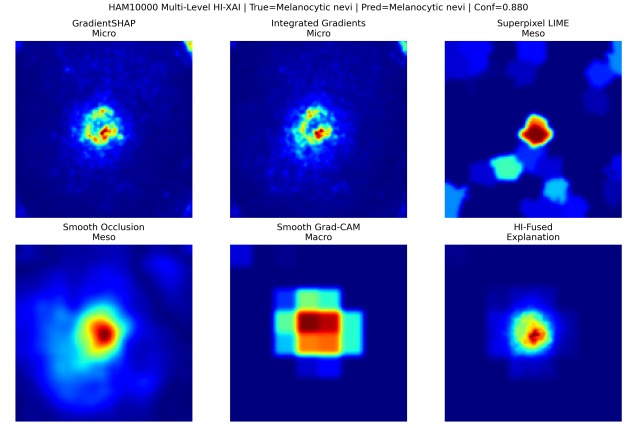


Fig. 6. XAI results of Melanocytic nevi skin cancer from HAM10000

The important finding is that the presence of diversity and instability of AI explainability cannot be avoided; however, these properties can be utilized to develop meaningful HI interpretability that can improve applications of XAI approaches (that is the proposed nEGXAI-V technique). XAI approaches provide diversified explainability across distinct data characteristics; however, these diversity properties can be utilized to support human interpretability of AI explainability.

This study also reveals some interesting properties (Fusion of XAI models and their uncertainty) that require further attention and in-depth research using the distinction between affirmation logic and non-affirmation logic that separates the XAI models with respect to scalability and complexity problems based on CNN-based models and data characteristics.

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